



Testing for unit roots, causality, cointegration, and efficiency: The case of the northwest US natural gas market

Yohannes Kebede Gebre-Mariam*

Office of the People's Counsel for the District of Columbia, 1133- 15th Street NW, Suite 500, Washington, DC 20005, USA

ARTICLE INFO

Article history:

Received 20 August 2010

Received in revised form

17 March 2011

Accepted 19 March 2011

Available online 22 April 2011

Keywords:

Natural gas

Causality

Efficiency

Cointegration

Energy

Time series

Market

ABSTRACT

The present study examined unit roots, causality, cointegration, and efficiency of the natural gas market using the Northwest US natural gas markets as a case study. The results indicate that the natural gas market prices are stationary after the first differencing, and that spot and futures natural gas prices move in a similar direction. Spot prices tend to exert significant influence on future prices for contracts with less than one year to maturity, but futures prices influence spot prices for contracts greater than one year to maturity. There are no arbitrage opportunities for daily market prices compared with weekly and monthly market price. Differences in regional market price are integrated of order one and characterized by long-run occurrence of the law of one price. Market prices in all market hubs move in similar direction with variance between spot and futures prices declining over time. Markets exhibit high risk premium for contracts with less than one year to maturity. The efficient market hypothesis holds true only for contracts with only about a month to maturity.

© 2011 Elsevier Ltd. All rights reserved.

1. Background

The North American continent is the largest market for natural gas in the world, with over 30% of worldwide consumption [1]. Within the United States economy, natural gas plays a major role because it accounts for almost a quarter of the total annual energy consumption. It is expected that US natural gas consumption will continue to increase due to a “relatively” competitive market structure, and environmental standards that encourage increased use or combustion of “cleaner” fuels for the generation of electricity. The increased demand for gas is expected to be met by the development of supplies from producing basins of the lower 48 states and imports from Canada.

The U.S. has a complex and extensive pipeline infrastructure for transporting natural gas from production areas to ultimate consumers. More than 160 intra- and inter-state natural gas pipeline companies operate more than 300,000 miles of transmission lines for delivery of gas throughout the lower 48 states [2]. These pipelines can deliver more than 148 billion cubic feet (Bcf) per day of natural gas. Hundreds of compressor stations and several storage facilities support these pipelines. Furthermore, about 1200 investor-owned

utilities (IOUs) (also called local distribution companies or “LDCs”) operate more than 1.2 million miles of pipeline infrastructure to deliver natural gas to customers [2].

It is clear that natural gas will continue to be vital to our economy. Therefore, it is important to study how natural gas markets operate, their efficiency and ways of improving the marketing system with the goal of finding ways to improve the efficiency of using natural gas. The present study will examine selected aspects of the natural gas market such as the relationships between spot and futures market, between spot prices at several regional markets, movement of prices across different markets, and the efficiency of the futures market.

The paper is organized as follows. Section 2 presents the problem investigated and the objectives of this study. Section 3 discusses natural gas spot and futures market. Section 4 presents data needs and empirical methodology. Section 5 discusses the results of empirical analysis. Section 6 presents the conclusions and recommendations, followed by references.

2. Problem statement

The North American natural gas market can be grouped into producing regions, consuming regions and market hubs. Pipelines interconnect these systems. The natural gas sector features the spot, futures, options, storage, capacity, and balancing markets. The market

* Tel.: +1 202 261 1415; fax: +1 202 727 1014.

E-mail address: ymariam@opc-dc.gov.

is considered liquid, well developed, and competitive. It is characterized by a high degree of volatility.

The natural gas industry is expected to grow. In fact, demand for natural gas is projected to increase at an average annual rate of 2.0 percent, from 22.8 to 33.8 trillion cubic feet between 2000 and 2020, primarily due to rapid growth in the demand for electricity generation [1]. Therefore, it is vital to study the market for this industry to improve its efficiency and conserve on this non-renewable clean source of energy.

Natural gas is one of the most actively traded commodities. Relatively high levels of liquidity and volatility, hence elevated market risk, characterize the natural gas market. Several factors influence the functioning of and risk in the natural gas market. These include production, transmission, distribution, storage, prices of competing fuels, demand, weather and market risk [3]. In recent years, natural gas has become one of the most volatile commodities traded on the New York Stock Exchange [4]. Natural gas futures variability and persistence of shocks tend to exhibit seasonality, thus affecting optimality of hedging [5].

Market risks primarily occur due to imbalances in the supply and demand reflected by variability in prices. Market risks can be anticipated and their impacts mitigated. Of the many types of natural gas market risks, price risk poses the greatest threat because the impacts are directly felt by customers and may have repercussive effects. Thus, it is important to identify and measure market risks with the objective of offering ways of mitigating the effect of increase in natural gas prices.

Despite the importance of this industry to the economy, little has been studied with respect to the linkages between the regional markets, and between futures and spot markets. Furthermore, little is known on how best to utilize price information from these markets to minimize risk.

Due to open access, companies can purchase gas at different market hubs and transport it to another location. The price premium that customers pay should, in a competitive market, only be the cost of transport, losses in fuel and transaction cost. Negotiating prices of long-term contracts could be much easier if there is prior knowledge that the long-run prices in one market are equal to prices at an alternate market, plus any transport charge that may exist. Furthermore, utilities, producers, and traders make use of prices in the futures market as well as in other regional markets to assess the prudence of gas purchase contracts.

In an efficient market, it is expected that prices in the futures market are unbiased or best estimators of forward spot market prices. Little evidence has been compiled so far to substantiate the efficiency of the natural gas futures market [6–8]. Availability of empirical evidence on efficiency of the natural gas market could help utilities, regulators, and traders in assessing strategies on how prices in futures market should or could be used to reduce costs and/or risks of natural gas to customers.

The present study intends to accomplish three major goals. First, the study intends to assess whether or not prices in spot and futures markets are linked. Second, the study investigates whether or not prices in various regional markets are linked, and operate under “the law of one price”. Third, the study tests for constancy of risk, rationality and efficiency of the futures market. Evidence on these and related issues can be useful to utilities, producers, traders, and regulators.

3. The natural gas spot and futures markets

3.1. The natural gas spot market

Deregulation of natural gas wellhead gas prices and open access to interstate pipelines created a structure in which LDCs, electric

utilities, industrial firms, commercial business, and gas trading companies purchase gas directly from sellers on a short-term basis. This structure constitutes the natural gas spot market. Due to a large number of participants, spot markets are expected to improve competition, liquidity, and efficiency in the pricing of natural gas.

Often, spot markets develop in regions where buyers and sellers are concentrated, such as at a pipeline interconnection near a large metropolitan area or at a major terminal in a gas-producing region. Several natural gas markets (hubs) connect to the lower 48 states of the United States and Canada. These include New York City Gate, Henry Hub (Henry), Chicago City Gate, Katy Hub, Texas, So. Calif. Border, AECO Hub (AECO) (Canada), ANR Oklahoma, CIG, Kern/Opal (Kern), Stanfield, Questar, Cheyenne Hub, NW WYM PL (Wyoming), Iroquois, Niagara, Sumas (Canada), Emerson (Viking), Dawn (Ontario), Kingsgate (Canada), Kern River Station, Malin, PG&E City gate. Among these, the New York Mercantile Exchange (NYMEX) prices are often taken as the barometer to discern movements in prices. These prices are helpful to assess incentives for drilling activities and to guide gas traders and utilities expectations of prices.

In the natural gas market, participants mainly trade natural gas under two types of contracts- physical and financial. These contracts are traded in different markets. Participants in these markets include producers, traders, suppliers, pipeline companies, and distribution utilities. The volume of trade in natural gas markets accounts for about 25% the total U.S. energy consumption [9]. Over the years, the natural gas market has become more complex and volatile. Consequently, natural gas market participants are unable to predict the future prices of natural gas, and therefore are exposed to price risk. The need to search for tools to minimize this risk led to the development of a financial gas market.

In the physical gas market, participants enter into physical gas contracts (sometimes referred to as cash gas contracts) for the physical delivery of natural gas. Gas contracts can be short term (or equal to one month), medium term (1–12 months) or long term (more than one year). The ability of participants to enter into these contracts has increased flexibility to form a portfolio of contracts that minimizes supply and price risks in both the long and the short run. A participant (producer, trader, supplier, and pipeline company or distribution firm) may purchase gas from the wholesale gas market for resale, or for end use in the retail gas market.

One of the most important features of natural gas is its ability to be stored. This attribute of natural gas has become an important element in improving the efficiency of the natural gas market. Storage ensures the availability of supplies during seasonal and short-term gas demand peaks. It also helps keep pipelines and distribution systems in physical balance. Furthermore, storage has increased the liquidity of the gas market, and thus contributed to improvements in the performance of commodity trading and management. Because natural gas can be stored, various lengths of physical contracts that offer different periods of consumption could be structured. The features of the natural gas market described above are expected to reveal the “market value” of the commodity traded through balancing supply with demand.

3.2. The natural gas futures market

Since the evolution of the commodity futures market in the 1750s leading up to the formation of the Chicago Board of Trade in 1848, the futures market has provided an important role in minimizing the price risk to suppliers and producers of various commodities [10]. The most important role of the futures market is that it permits market participants to hedge the risk of price volatility. Unlike the spot market, futures market trading provides

a way to establish a form of price knowledge leading to continuous price discovery. Futures prices reflect not only current cash prices, but also expectations of future prices and general economic factors. When markets are not undistorted or when they are considered efficient, the price signals generated in the futures market should accurately reflect the future spot price of natural gas. It is only under this kind of situation that producers and suppliers can effectively hedge the risk of future price changes.

The New York Mercantile Exchange (NYMEX) launched the world's first natural gas futures contract in April 1990 [11]. The NYMEX delivery point at Henry Hub, located in Earth, Louisiana, was chosen as a benchmark to set gas prices. Since 1992, the exchange launched options on natural gas futures, derivatives and other hedging mechanisms that give market participants additional flexibility in managing their market risk.

Until the beginning of the 1990s, the natural gas market was stable and controlled, and dominated by long-term contracts. Since then, the market has changed to a dynamic and competitive market with an active spot market and effective use of short and medium-term contracts. The natural gas futures market added the option of managing price risk. Today, the natural gas futures market and spot markets involve diverse participants and market products. Since participants engage in both markets, prices in the two markets tend to move in similar directions.

4. Data and empirical methodology

4.1. Data

The increasing role of natural gas as the major source of energy requires an increased understanding of its market structures, efficiency, and integration. This study investigates several attributes of the natural gas market. First, the study examines causal relationships between spot and futures markets. Second, the study investigates cointegration between spot and futures markets, and between spot prices at various regional markets. Lastly, the study assesses the efficiency or rationality of the natural gas futures market. The statistical analyses of market prices that are required to achieve the objectives of this study are described below.

The study uses natural gas prices reported in *Gas Daily*. *Gas Daily* reports daily wholesale spot prices at several trading locations throughout the U.S. natural gas transmission network. *Gas Daily* surveys daily those companies that trade natural gas to obtain a representative number of deals at each pricing point. All prices are quoted in dollars per million British Thermal Units (\$/MMBtu) or per Decatherm (\$/Dth).

The present study conducts several statistical analyses of natural gas prices. Spot price data for many regional markets and futures data from NYMEX are used in this study. The spot price data covers the period of November 1996 to December 2004.¹ Lack of data on futures prices necessitated the use of a relatively shorter duration of data, which covers the period of 1999 to 2004. All futures prices were divided into groups according to their distance from delivery or maturity and the data were aggregated by delivery months.

4.2. Empirical methodology

4.2.1. Tests for stationarity

Similar to other economic time series data, prices of natural gas are believed to exhibit non-stationarity [6,12,13]. Testing for this characteristic of prices is necessary before performing other tests

because most conventional statistical tests assume that time-dependent variables exhibit stationary behavior.² Therefore, it is essential to implement modeling and testing procedures for unit root to ascertain the nature of movements or the long-run relationship between spot and futures prices, as well as the property of prices in various regional market hubs.

It is widely recognized that most economic time series data exhibits a monotonically increasing trend. It means the error terms underlying the distribution of these series may not be stationary. Statistical analysis of these kinds of data, without removing non-stationarity, results in what is termed as “spurious” regression. Thus, it is imperative to examine stationarity of the individual time series prior to performing tests of causality or cointegration because the asymptotic distributions of causality tests are sensitive to the presence of unit roots and trends. If correction is not made to the unit-root process, the result may show evidence of a trend when none is present or the t-test will tend to indicate a relationship between the variables when none is present. If a series is behaving like a random walk (regression is $P_t = \alpha + \beta P_{t-1}$), the estimate of the slope coefficient is biased downwards resulting in rejection of the null hypothesis that $\beta = 1$.

In the natural gas market, several studies fail to reject the null hypothesis of a unit root and therefore conclude that gas prices are non-stationary [14–17]. Moreover, these studies have shown that natural gas prices are stationary at first difference.

Let us take an autoregressive process of order 1 (AR(1)):

$$\Delta P_t = \mu + \beta P_{t-1} + \varepsilon_t \quad (1)$$

where μ and β are parameters, and the ε_t is assumed to be independently and identically distributed with zero mean and equal variance. If $-1 < \beta < 1$, the process is stationary. If $\beta = 1$, the equation represents a process that is a random walk with a drift. This process characterizes most economic time series data. Therefore, the appropriate statistical test for random walk process is $\beta = 1$. This constitutes a test for unit-root and determines whether the series is non-stationary or not. The series must be demonstrated to be stationary as a precondition prior to performing other statistical analyses. The present study will use two commonly used methods to test for the existence of unit roots, namely the Phillips–Perron (PP) and the augmented Dickey–Fuller (ADF) tests [18,19].

The present study will employ the Augmented Dickey–Fuller (ADF) and the Phillip–Perron (PP) test in order to assess whether or not the price series from several regional markets exhibit stationarity. The test for unit root was performed under two scenarios: without a trend variable and with a trend variable. The reason for testing with and without a trend is that the price variables in some markets may show a sustained positive trend in the level of the series compared with other markets. Therefore, to minimize errors of inappropriate test of stationarity due to a trending variable, it is imperative to test for unit root with and without assuming time trend [6,9,14–17,20].

4.2.2. Test for causality

Doane and Spulber [8] employed Granger causality to assess U.S. gas market integration. The study found Granger causality between only one of 20 pairs of gas prices from five regions prior to open access. After open access, however, all 20 tests produced evidence of instantaneous bi-directional Granger causality. It is argued that a gap of one month may not allow all arbitrage opportunities to be

¹ Lack of funding to collect additional data limited the volume of data to be used in this study. Future study could extend the time period to present time.

² If the values of a variable exhibit a constant mean and standard deviation over time, then the variable is stationary.

exploited. Similarly, Chinn et al. [12] found that natural gas futures were less successful or not statistically significant in predicting future spot prices at 3 and 6 months horizons. Data that covered a 12-month horizon, however, showed evidence of market efficiency (predict future spot prices) but explained only 21% of variation (r -square) [12].

One of objective of the present study is to assess the nature of the relationship between spot prices at various regional markets, and between the futures and spot prices of natural gas. In order to accomplish this task, the following approach is adopted. Unit-root tests are performed on the price series in order to determine the existence of non-stationarity. Once this test is completed, tests for causality are implemented.

In performing causality testing, bivariate regressions are implemented. The Henry Hub market is the most liquid and used by most market participants compared to others regional markets. Therefore, causal relationships between spot prices observed at Henry Hub and regional markets are examined. In addition, bivariate causal relationships between futures and spot prices are implemented.

The regressions used in assessing bivariate causal relationships [12,21,22] can be specified as:

$$Y_t = \alpha_0 + \alpha_1 Y_{t-1} + \alpha_2 Y_{t-2} + \dots + \alpha_n Y_{t-n} + \beta_1 X_{t-1} + \beta_2 X_{t-2} + \dots + \beta_n X_{t-n} + \varepsilon_t, \quad t = 1, 2, \dots, m \quad (2)$$

$$X_t = \alpha_0 + \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \dots + \alpha_n X_{t-n} + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \dots + \beta_n Y_{t-n} + \eta_t, \quad t = 1, 2, \dots, m \quad (3)$$

In this equation, y and x represent markets y and x or spot and future prices, α and β , respectively represent the intercept and unknown parameters, and ε and η are error terms.

The bivariate regressions in equations (2) and (3) are implemented for all possible pairs of the series. The test statistic to determine whether (i) futures prices cause movements in spot price or vice versa, and (ii) whether or not prices in one regional market causes movements in prices of another market is given by F -statistics (also referred as Wald statistic). This statistic is used to test the joint hypothesis of the following type:

$$\beta_1 = \beta_2 = \dots = \beta_n = 0 \quad (4)$$

The test in equation (4) is performed for each equation.

Several studies have argued that the natural gas market is competitive and thus there is an instantaneous relationship between prices across regional markets, as well as between future and spot prices [6,7,14–16,23]. However, the relationship between prices at several regional markets, and between spot and futures may not be instantaneous but observed over a longer time. Numerous studies assert the futures market causes fluctuations in spot market prices [24–27]. Others claim that the direction of causality runs both ways, spot to futures and futures to spot ($\Leftrightarrow$). Knowledge on the direction of causality and how early in contract maturation date (quote date) that significant causation is observed would be important to hedgers, managers and policy makers in the energy sector.

4.2.3. Tests for cointegration

Engle and Granger [28] argued that if a linear combination of two or more non-stationary series can become stationary, then the stationary linear combination is called the cointegrating equation and may be interpreted as a long-run equilibrium relationship among the variables. In other words, if markets for natural gas are cointegrated, then it is likely that market prices converge to the law of one price (LOP), excluding transportation and distribution charges or arbitrage. Cointegration tests are performed to assess

evidence on the existence of long-term statistical relationships between future and spot natural gas prices observed across different regional markets.

Several studies have found that natural gas prices are cointegrated [6,8,14–17,20]. Cointegration of prices is evidence that natural gas market participants are able to achieve geographic arbitrage between locations on the pipeline network. The findings from these studies indicate that gas markets have become increasingly integrated after deregulation. Furthermore, it was found that open access broadened the geographic scope of the spot market and showed cointegration [20,29].

There are several natural gas markets in the U.S. Most regional markets seem to follow or track prices observed at Henry Hub and NYMEX. A recent study by Serletis and Rangel-Ruiz [30] found that the North American natural gas prices in the deregulated period are largely defined by the U.S. Henry Hub price trends. The Henry Hub market is also the most highly traded or liquid hub. Therefore, the present study intends to test cointegration of this market with other regional markets.

Let us assume that the study investigates the co-movement of prices in two markets, “ x ” and “ y ”. Let the price of gas in market “ x ”, denoted as P_x , be influenced by the price in market “ y ”, denoted as P_y . The extent of the influence can be investigated through an equation that models the market price at location “ x ” as a function of some variant of the transportation/transaction costs, denoted as α_{xy} , and some fraction β of the price in market “ y ”. This can be expressed as,

$$P_{xt} = \alpha_{xyt} + \beta Y_{yt} + \eta_{xt} \quad (5)$$

where α and β are coefficients, and η is white noise. The criterion to test for cointegration is that prices in the two markets should not drift apart without limit. That is, the error term must be stationary. It implies that the relationship between P_x and P_y will return to the equilibrium or long run equation $P_{xt} = \alpha_{xyt} + \beta P_{yt}$. More formally, if a vector of time series is $I(d)$ but a linear combination is integrated to a lower order, the time series is said to be cointegrated. If $\beta > 1$, then the two price series drift apart and the series are not considered cointegrated. P_x and P_y are said to be cointegrated if there is a weighted sum (linear combination) of the series, called the “cointegrating vector” and denoted by “ z ”, is stationary. In other words, P_x and P_y are cointegrated in $P_x, P_y \sim I(1)$ but there exists β such that $Z = P_x - \beta P_y \sim I(0)$.

Based on the discussion above, the LOP is a particular case where β is equal to one. This would suggest that the markets are integrated. If markets are integrated, then gas can flow or freely arbitrated between two or more market locations. In addition, prices are free to adjust so that the price difference would be the transportation/transaction costs. Determination as to whether the LOP holds can be made by assessing whether or not β is equal to, or very close to, a value of one. The closer β is to one, the more integrated are the two market locations.

In this study, vector autoregressive model (VAR) based cointegration test using the methodology developed in Johansen [31,32] is used. Given markets “ x ” and “ y ” in equation (5), a VAR of order p can be written as:

$$Y_t = A_1 Y_{t-1} + \dots + A_p Y_{t-p} + B X_t + \varepsilon_t \quad (6)$$

where A and B are matrices of coefficients to be estimated, Y is a k -vector of non-stationary endogenous variable of $I(1)$, X is a d -vector of exogenous variables, and ε is a vector of innovations.

When formulating models with time series variables, one must be concerned with the stability (stationarity) properties of the variables. That is, it is important to determine whether (i) a variable drifts from its mean, and (ii) there is a tendency toward long-term

equilibrium or not. One method of testing stationary property of a variable is to undertake cointegration tests.

According to Phillips [33], if a set of variables found to be integrated of order one or $I(1)$ but not cointegrated, any regression involving the levels of these variables is spurious. This implies that only cointegrated variables can be used in a model that is expressed in levels of the variables. However, if the variables are not cointegrated, any model involving these variables should be stated in its first difference. As indicated in previous section, tests for the presence of unit root indicated that the price series are stationary in the first difference. Thus, they are considered to be $I(1)$ and cointegrated. Therefore, a vector error correction (VEC) model was formulated and estimated.

4.2.4. Testing for efficiency or rationality of futures market

In an analysis of the linkages between spot and futures price volatility, it is important to examine the extent to which futures market prices accurately reflect the available information that bears on the spot price. In general, futures markets are claimed to be efficient when the futures prices today contain all relevant information to forecast tomorrow's spot price [34].

The literature that studied the efficiency of natural gas futures markets and their role in developing a hedging strategy is scant [6,8,14,35,36]. Chinn et al., [12] found that futures prices are unbiased predictors of crude oil, gasoline and heating oil prices, but not of natural gas prices. The present research investigates whether or not the market efficiency hypothesis holds true for the natural gas futures market.

4.2.4.1. Empirical specification. Let S_T and F_{T-1} represent spot and futures prices that correspond to the spot and futures price at time "T", respectively. Allowing for a stochastic rather than a deterministic equivalence, and on the assumption of perfect knowledge on determinants of price movements, the condition that needs to be satisfied for the weak form of efficiency is:

$$E(S_T - F_{T-i,T}|O_{T-i}) = 0 \quad (7)$$

where E is the expectation operator, S_T is spot price at time T , $F_{T-i,T}$ is futures price at time $T-i$ for delivery at time T , and O_{T-i} is information set at time $T-i$. Equation (7) says that the futures price $F_{T-i,T}$ quoted at time $T-i$ for delivery at time T is an unbiased estimate of the spot price S_T at the delivery date given the information set available at time $T-i$.

The testable implication of (7) is that the forecast error should be uncorrelated with information in O_{T-i} . One way of doing this is to regress $(S_T - F_{T-i,T})$ on variables contained in O_{T-i} . The null hypothesis of rationality is rejected whenever the coefficient of any one of these variables is significantly different from zero. However, data pertaining to O_{T-i} is not available. Hence, it cannot be tested in this study.

Another commonly used alternative is a regression test. Consider a sequence of futures prices $F_{t,T}, F_{t+1,T}, \dots, F_{T-1,T}$ quoted at consecutive trading dates for the same delivery date (T). If the market is efficient, then the regression equations take the following form:

$$S_T = \alpha_i + \beta_i F_{T-i,T} + \varepsilon_T, \quad i = 1, 2, \dots, T-1 \quad (8)$$

Under the assumption of an efficient market, α_i in equation (8) represents transmission and transaction costs (see also equation (5)). In dealing with futures and spot prices, the transaction costs include the cost incurred to gather and process information as well as the degree of risk averseness (risk premium) for using futures prices as unbiased estimates of spot prices. In the past, the

literature assumed that α_i represents only transmission costs, and values of this parameter that are greater than tariff transportation rates are assumed to be indicators of market inefficiency, and existence of arbitrage opportunities [6–8].

Let us assume that α_i is composed of two components, a risk premium (γ) and transportation (λ) cost. Let us also assume that the regulated transmission cost is θ . The efficiency hypotheses in (8) implies $\alpha_i = \theta$, and $\beta_i = 1$ for all i , where the ε_T 's are serially uncorrelated and i denotes the number of weeks before delivery at which the futures price is quoted. If S and F are cointegrated, then ε_t will be stationary. The accumulation of information with the passage of time also suggests that the value of R^2 should increase monotonically with $T-i$. Hence, the prices of near futures contracts should better estimate delivery spot prices than the more distant futures contracts. The intercept term, in part, incorporates an ex-post measure of the risk premium. That is, even if the market is efficient, the spot and futures prices could differ by the amount of transportation and distribution charges and the risk premium. If the market is efficient, futures prices will be the unbiased estimator of spot price only up to the cost of distribution and transmission.

A finding that $\alpha \geq \theta$ and $\beta = 1$ in equation (8) could imply the existence of risk aversion. However, the expectation remains rational unless $\beta \neq 1$. The null hypothesis $\alpha = 0$ and $\beta = 1$, constitutes a joint test of (constant) risk neutrality and rationality, subject to qualifications such as no correlation of the error term with the information set. The formulation to be used in this study is in logs rather than in absolutes, since a risk premium constancy is more easily defended as a proportion [37]. That is, if the market is efficient then the

$$\lim_{T-i \rightarrow T} \frac{S_T}{F_{T-i,T}} = 1 \quad (9)$$

In other words, the ratio should be approximately equal to one for efficient markets. A linear relationship of this form of proportion can be derived through logarithmic transformation, (i.e., $\log S_T = \log F_{T-i,T}$). Equation (8) is used to estimate efficiency of the futures market. It also demonstrates the long-run relationship between the two price series.

The ex-post risk premium (α_i) can be represented as $\alpha_i = S_T - \beta_i F_{T-i,T}$ [24].³ Depending on the expectation of participants and their risk taking behavior, α_i can take negative and positive values. Risk-neutrality, or a value of $\alpha = 0$, indicates the convergence of spot and futures prices. Expectation of spot price greater than futures prices, $\alpha > 0$, imply risk aversion. A negative value, $\alpha < 0$, indicates risk-taking behavior. However, it should be noted that a negative intercept term does not imply that market participants are risk takers unless supplemented with appropriate information necessary to characterize risk-taking behavior of decision-makers or market participants. Theoretically, as contracts approach maturity (i.e., $i \rightarrow T$), decision makers accumulate information and knowledge of the market place. As a result, the intercept term (risk premium) is expected to decline.

In the equation $S_T = \alpha_i + \beta_i F_{T-i,T}$, a rejection of the hypothesis that $\beta_i = 1$ or rationality of futures market imply that the risk premium (α) will be non-stationary. Furthermore, it implies the futures prices, hence market information embodied in it, are not unbiased predictors of the forward spot markets.

³ Ex-post risk premium is based on historical volatility of prices, while ex-ante risk premium is computed from expected prices movements. These premiums move in opposite direction because prices with high-risk premium (ex-post) thus volatile will have low ex-ante risk premium, and vice versa.

5. Results of empirical analysis

5.1. Stationarity/nonstationarity of price series

The results of unit root tests are presented in Table 1. In all cases, the results indicate that prices in all markets show the presence of unit root. However, testing for unit root after differencing resulted in rejection of the null hypothesis of the presence of unit root (non-stationarity) in all cases. That is, the value of the test statistics was greater than MacKinnon's critical values [38].

Sensitivity runs of unit root tests for different lags did not show significantly different results. The finding, therefore, confirm that the price series is stationary in the first difference. That is, natural gas prices are integrated of order one, $I(1)$. Once the price series are found to be $I(1)$, the next steps in the analysis of natural gas prices will test for causality, co-integration and market efficiency.

5.2. Results of causality tests

5.2.1. Tests for causality between spot and futures market

Analysis of the relationship between futures and spot prices is dictated by the fact that data on futures prices was available by delivery month. Therefore, the data to test causal relationship between spot and futures prices was aggregated by month.

The results of causality analysis between forward and spot prices are presented in Table 2.

The impact of spot price on futures prices is statistically significant for contracts that are about 6 months to maturity. Over a one year time period, spot markets significantly affect movements in forward price in about 26 out of 52 (50%) of the cases. Furthermore, spot prices exert statistically significant impact on forward prices that are up to 4 months to maturity. This impact reduced to only about 5% for contract with 1–2 year to maturity.

Over a period of two years, however, spot prices caused statistically significant change in forward price in only one-third (30%) of the cases. On the other hand, futures prices exert statistically significant impacts on spot market with contracts of up to three months to maturity. Furthermore, future prices exert statistically significant influence on contracts that are within 1 year and three (15 months) months to two year (24 months) before maturity. In fact, future price exert significant impact on spot price for contracts with 1–2 years before maturity in 50% of the time (27 weeks out of 52 weeks). It seems that spot price significantly influences forward prices in the first 12 months. However, futures prices with contract maturity of 12 months and beyond appears to be a significant driver of spot prices.

The impact of forward price on spot price and vice versa seems to change by time period and hence by the accuracy of available market information. For prices series observed within a year, spot market prices seem to exert greater influence compared to forward price. However, it appears that there is a bi-directional casual relationship (spot to forward and vice versa) during the first three months before contract maturity. As the accuracy of market information declines (contract maturity period increased), the impact of forward prices on spot prices seems to be greater than the impact of spot prices on forward prices. The implications of this finding is that if an individual or company wishes to utilize information on the movement of futures prices to enter into hedging, it is recommended to examine price movements or the relationships of not only three months to maturity but also those that are 15–24 months before maturity of contracts.

One of the distinguishing features of natural gas as a commodity compared with other storable commodities is that natural gas cannot be sold or transported in smaller chunks or quantities. These characteristics of natural gas could affect gains from economies of scale. The cost of storage for natural gas is relatively expensive compared with other commodities. Natural gas can be transported over a delimited distance (e.g., along the northwest and western North American corridor). Finally, natural gas comes as a secondary product of crude oil exploration. Because of these product specific characteristics and related factors, the impact of futures price on spot price may be less than the impact of spot on futures prices. In other words, in the short run spot prices seem to have the greatest impact on futures prices but long term future price exert greater influence on spot prices.

5.2.2. Tests for causality between regional markets

The result of causality between regional markets is presented in Table 3. The results of causality tests between regional markets exhibit a bi-directional or two-way causal relationship in most markets. This means that there is an instantaneous response of price changes across markets, thus limiting the possibility of arbitrage. The results also show that markets that are directly connected with major sources of supply tend to exhibit a statistically significant relationship compared with markets that are not connected or are not in close proximity to sources of supply. For instance, Canadian markets such as AECO, Henry Hub and Sumas are located closer to source of supply (Canadian) than other regional markets.

The findings indicate that movements of prices in markets located close to supply sources tend to influence prices in other regional markets. Prices in Canadian markets seem to exert

Table 1
Unit root test on natural gas spot prices and futures prices, 1996–2004.

Variables	Augmented Dickey–Fuller test statistic				Phillips–Perron test statistic			
	With trend ^a		Without trend ^a		Without trend ^a		With trend ^a	
	Critical values	Prob.	Critical values	Prob.	Critical values	Prob.*	Critical values	Prob.
Monthly forward	−2.890	0.053**	−3.195	0.096**	−1.774	0.389	−1.972	0.604
AECO	−1.999	0.287	−2.047	0.564	−2.159	0.223	−2.239	0.460
Henry	−2.120	0.238	−2.161	0.502	−2.120	0.238	−2.331	0.411
Kern–Opal	−1.773	0.390	−1.830	0.677	−1.976	0.296	−2.047	0.564
Kingsgate	−3.226	0.024*	−3.200	0.095**	−3.285	0.020*	−3.263	0.083**
Sumas	−3.220	0.024*	−3.193	0.096**	−3.289	0.020*	−3.266	0.083**
Malin	−3.140	0.029*	−3.112	0.114	−3.140	0.029*	−3.112	0.114
WYOMING	−1.776	0.388	−1.831	0.677	−1.982	0.294	−2.051	0.561
Stanfield	−3.257	0.022*	−3.229	0.089**	−3.330	0.018*	−3.306	0.076**

*, ** – Significant at 5 and 10% level of significance respectively.

^a MacKinnon's critical values for 5% and 10% (without trend regression) are −2.914 and −2.595, and for with trend −3.491 and −3.174 respectively.

Table 2Results of testing for causality between spot prices henry hub and futures prices from New York Mercantile Exchange (NYMEX).^a

Direction of causality	Prob.	Direction of causality	Prob.	Direction of causality	Prob.
WK1 → Henry hub	0.0128*	WK18 → Henry Hub	0.9502	WK35 → Henry Hub	0.5493
Henry Hub → WK1	0.000*	Henry Hub → WK18	0.0078*	Henry Hub → WK35	0.4035
WK2 → Henry Hub	0.1606	WK19 → Henry Hub	0.9657	WK36 → Henry Hub	0.6074
Henry Hub → WK2	0.000*	Henry Hub → WK19	0.0044*	Henry Hub → WK36	0.6195
WK3 → Henry Hub	0.157	WK20 → Henry Hub	0.8149	WK37 → Henry Hub	0.737
Henry Hub → WK3	0.000*	Henry Hub → WK20	0.0072*	Henry Hub → WK37	0.7157
WK4 → Henry Hub	0.1563	WK21 → Henry Hub	0.2208	WK38 → Henry Hub	0.5887
Henry Hub → WK4	0.000*	Henry Hub → WK21	0.0121*	Henry Hub → WK38	0.4419
WK5 → Henry Hub	0.0178*	WK22 → Henry Hub	0.050*	WK39 → Henry Hub	0.4967
Henry Hub → WK5	0.000*	Henry Hub → WK22	0.0346*	Henry Hub → WK39	0.252
WK6 → Henry Hub	0.2966	WK23 → Henry Hub	0.0664**	WK40 → Henry Hub	0.4534
Henry Hub → WK6	0.000*	Henry Hub → WK23	0.0128*	Henry Hub → WK40	0.1562
WK7 → Henry Hub	0.3895	WK24 → Henry Hub	0.1448	WK41 → Henry Hub	0.4672
Henry Hub → WK7	0.000*	Henry Hub → WK24	0.0055*	Henry Hub → WK41	0.5554
WK8 → Henry Hub	0.1309	WK25 → Henry Hub	0.2749	WK42 → Henry Hub	0.6415
Henry Hub → WK8	0.000*	Henry Hub → WK25	0.0477*	Henry Hub → WK42	0.7619
WK9 → Henry Hub	0.082**	WK26 → Henry Hub	0.5198	WK43 → Henry Hub	0.6605
Henry Hub → WK9	0.000*	Henry Hub → WK26	0.1139	Henry Hub → WK43	0.9661
WK10 → Henry Hub	0.0324*	WK27 → Henry Hub	0.5572	WK44 → Henry Hub	0.6529
Henry Hub → WK10	0.000*	Henry Hub → WK27	0.2003	Henry Hub → WK44	0.9654
WK11 → Henry Hub	0.0754**	WK28 → Henry Hub	0.6187	WK45 → Henry Hub	0.658
Henry Hub → WK11	0.000*	Henry Hub → WK28	0.2456	Henry Hub → WK45	0.9501
WK12 → Henry Hub	0.3376	WK29 → Henry Hub	0.5279	WK46 → Henry Hub	0.6324
Henry Hub → WK12	0.000*	Henry Hub → WK29	0.4977	Henry Hub → WK46	0.9033
WK13 → Henry Hub	0.5464	WK30 → Henry Hub	0.3808	WK47 → Henry Hub	0.6414
Henry Hub → WK13	0.000*	Henry Hub → WK30	0.4892	Henry Hub → WK47	0.8972
WK14 → Henry Hub	0.5916	WK31 → Henry Hub	0.2662	WK48 → Henry Hub	0.4584
Henry Hub → WK14	0.002*	Henry Hub → WK31	0.2425	Henry Hub → WK48	0.5957
WK15 → Henry Hub	0.4491	WK32 → Henry Hub	0.3657	WK49 → Henry Hub	0.343
Henry Hub → WK15	0.0032*	Henry Hub → WK32	0.1706	Henry Hub → WK49	0.7051
WK16 → Henry Hub	0.7441	WK33 → Henry Hub	0.6637	WK50 → Henry Hub	0.5002
Henry Hub → WK16	0.002*	Henry Hub → WK33	0.2946	Henry Hub → WK50	0.8261
WK17 → Henry Hub	0.9087	WK34 → Henry Hub	0.5136	WK51 → Henry Hub	0.7328
Henry Hub → WK17	0.0034*	Henry Hub → WK34	0.3902	Henry Hub → WK51	0.9736
WK52 → Henry hub	0.9398	WK70 → Henry hub	0.0364*	WK88 → Henry hub	0.0612**
Henry hub → WK52	0.996	Henry hub → WK70	0.4084	Henry hub → WK88	0.3154
WK53 → Henry hub	0.9953	WK71 → Henry hub	0.0218*	WK89 → Henry hub	0.1067**
Henry hub → WK53	0.9257	Henry hub → WK71	0.5892	Henry hub → WK89	0.5177
WK54 → Henry hub	0.4936	WK72 → Henry hub	0.0775**	WK90 → Henry hub	0.231
Henry hub → WK54	0.9232	Henry hub → WK72	0.6669	Henry hub → WK90	0.4976
WK55 → Henry hub	0.6418	WK73 → Henry hub	0.0859**	WK91 → Henry hub	0.1178
Henry hub → WK55	0.816	Henry hub → WK73	0.3428	Henry hub → WK91	0.3667
WK56 → Henry hub	0.6192	WK74 → Henry hub	0.052**	WK92 → Henry hub	0.2374
Henry hub → WK56	0.6926	Henry hub → WK74	0.1828	Henry hub → WK92	0.3762
WK57 → Henry hub	0.5979	WK75 → Henry hub	0.0814**	WK93 → Henry hub	0.2452
Henry hub → WK57	0.4064	Henry hub → WK75	0.195	Henry hub → WK93	0.3712
WK58 → Henry hub	0.5084	WK76 → Henry hub	0.0665**	WK94 → Henry hub	0.1316
Henry hub → WK58	0.4223	Henry hub → WK76	0.1448	Henry hub → WK94	0.0967**
WK59 → Henry hub	0.5826	WK77 → Henry hub	0.1485	WK95 → Henry hub	0.2511
Henry hub → WK59	0.4248	Henry hub → WK77	0.1738	Henry hub → WK95	0.1615
WK60 → Henry hub	0.5877	WK78 → Henry hub	0.058**	WK96 → Henry hub	0.1888
Henry hub → WK60	0.5382	Henry hub → WK78	0.2452	Henry hub → WK96	0.0601**
WK61 → Henry hub	0.5663	WK79 → Henry hub	0.0978**	WK97 → Henry hub	0.0982**
Henry hub → WK61	0.5682	Henry hub → WK79	0.2021	Henry hub → WK97	0.0159*
WK62 → Henry hub	0.4963	WK80 → Henry hub	0.0504*	WK98 → Henry hub	0.0693**
Henry hub → WK62	0.4181	Henry hub → WK80	0.3535	Henry hub → WK98	0.0548**
WK63 → Henry hub	0.4966	WK81 → Henry hub	0.0283*	WK99 → Henry hub	0.090**
Henry hub → WK63	0.6926	Henry hub → WK81	0.3518	Henry hub → WK99	0.3714
WK64 → Henry hub	0.2647	WK82 → Henry hub	0.0882**	WK100 → Henry hub	0.1738
Henry hub → WK64	0.5703	Henry hub → WK82	0.4169	Henry hub → WK100	0.3199
WK65 → Henry hub	0.1331	WK83 → Henry hub	0.1194	WK101 → Henry hub	0.0734**
Henry hub → WK65	0.5725	Henry hub → WK83	0.2667	Henry hub → WK101	0.44
WK66 → Henry hub	0.1575	WK84 → Henry hub	0.0506*	WK102 → Henry hub	0.039*
Henry hub → WK66	0.2988	Henry hub → WK84	0.153	Henry hub → WK102	0.3277
WK67 → Henry hub	0.0407*	WK85 → Henry hub	0.0802**	WK103 → Henry hub	0.0771**
Henry hub → WK67	0.5196	Henry hub → WK85	0.173	Henry hub → WK103	0.2598
WK68 → Henry hub	0.0387*	WK86 → Henry hub	0.126	WK104 → Henry hub	0.048*
Henry hub → WK68	0.5102	Henry hub → WK86	0.0877**	Henry hub → WK104	0.2311
WK69 → Henry hub	0.0615**	WK87 → Henry hub	0.1173	→	
Henry hub → WK69	0.2148	Henry hub → WK87	0.0779**	→	

*, ** Significant at 5% and 10% respectively.

^a Where x75, X74... X1, refers to 75,74,...1 week before maturity of contracts.

Table 3
Testing for pair-wise Granger causality.

Direction of causality	Daily	Weekly	Monthly	Direction of causality	Daily	Weekly	Monthly
Henry Hub → Aeco	0.0000*	0.2317	0.594	Kern → Malin	0.0000*	0.0013*	0.2703
Aeco → Henry Hub	0.0000*	0.0000*	0.9625	Wyoming → Kern	0.0000*	0.2938	0.3488
Kern → Aeco	0.3816	0.5582	0.8553	Kern → Wyoming	0.0000*	0.0055*	0.3041
Aeco → Kern	0.0000*	0.0347*	0.9031	Stanfield → Kern	0.0000*	0.0000*	0.0011*
Kingsgate → Aeco	0.0000*	0.0000*	0.0036*	Kern → Stanfield	0.0000*	0.0180*	0.1427
Aeco → Kingsgate	0.0000*	0.0123*	0.0881**	Sumas → Kern	0.0000*	0.0000*	0.0012*
Malin → Aeco	0.0000*	0.0010*	0.0132*	Kern → Sumas	0.0000*	0.0149*	0.1434
Aeco → Malin	0.0000*	0.0036*	0.3965	Malin → Kingsgate	0.0000*	0.0006*	0.3481
Wyoming → Aeco	0.0885**	0.6971	0.8533	Kingsgate → Malin	0.0000*	0.0016*	0.0843**
Aeco → Wyoming	0.0000*	0.0118*	0.8629	Wyoming → Kingsgate	0.0000*	0.0098*	0.1134
Stanfield → Aeco	0.0000*	0.0000*	0.0031*	Kingsgate → Wyoming	0.0000*	0.0000*	0.0007*
Aeco → Stanfield	0.0000*	0.0269*	0.0759**	Stanfield → Kingsgate	0.0000*	0.0000*	0.139
Sumas → Aeco	0.0000*	0.0000*	0.0026*	Kingsgate → Stanfield	0.0000*	0.0000*	0.3179
Aeco → Sumas	0.0000*	0.0147*	0.0843**	Sumas → Kingsgate	0.0000*	0.0000*	0.1005**
Kern → Henry Hub	0.0000*	0.0032*	0.9667	Kingsgate → Sumas	0.0000*	0.0001*	0.2318
Henry Hub → Kern	0.0441**	0.9899	0.7593	Wyoming → Malin	0.0000*	0.0012*	0.2617
Kingsgate → Henry Hub	0.0000*	0.0000*	0.0003*	Malin → Wyoming	0.0000*	0.0004*	0.0046*
Henry Hub → Kingsgate	0.0000*	0.0800**	0.1181	Stanfield → Malin	0.0008*	0.0001*	0.0162*
Malin → Henry Hub	0.0000*	0.0001*	0.0023*	Malin → Stanfield	0.0000*	0.0106*	0.0870**
Henry Hub → Malin	0.0000*	0.0029*	0.4638	Sumas → Malin	0.0000*	0.0034*	0.0096*
Wyoming → Henry Hub	0.0000*	0.0049*	0.94	Malin → Sumas	0.0000*	0.1776	0.0605**
Henry Hub → Wyoming	0.2518	0.9371	0.6878	Stanfield → Wyoming	0.0000*	0.0000*	0.0012*
Stanfield → Henry Hub	0.0000*	0.0000*	0.0002*	Wyoming → Stanfield	0.0000*	0.0154*	0.1385
Henry Hub → Stanfield	0.0000*	0.0785**	0.1262	Sumas → Wyoming	0.0000*	0.0000*	0.0013*
Sumas → Henry Hub	0.0000*	0.0000*	0.0002*	Wyoming → Sumas	0.0000*	0.0130*	0.1388
Henry Hub → Sumas	0.0000*	0.0854**	0.1425	Sumas → Stanfield	0.0000*	0.5377	0.3824
Kingsgate → Kern	0.0000*	0.0000*	0.0006*	Stanfield → Sumas	0.0000*	0.0684**	0.3729
Kern → Kingsgate	0.0000*	0.0115*	0.1168	→			
Malin → Kern	0.0000*	0.0019*	0.0044*	→			

*** – Significant at 5% and 10% level significance, respectively.

relatively greater impact on the movement of prices in U.S. markets. The results also show that there exist a bi-directional causal relationships among all regional markets when daily prices are used to study market behavior compared with a situation where weekly and monthly prices are used. This means that opportunities for arbitrage or profit are limited from using daily prices compared with using weekly or monthly prices.

The results of causality testing indicate that when weekly price are used, not only the direction of causality become insignificant or only unidirectional but also the magnitude of the impacts changed compared with a situation where daily prices are used. Moreover, when monthly price data is used, price relationships that were insignificant become significant (e.g., Kern and AECO, and Sumas and Malin). The implication of this finding is that the extent to which prices at various regional markets are used in relation to prices at Henry Hub may produce varying results depending on whether daily, weekly or monthly prices are used. Knowledge of these market relationships and their behavior over time could be used to guide the purchase of least cost resources or profit from price differentials at different hubs or basins.

In summary, it is plausible to conclude that the various regional markets included in this study show statistically significant bi-directional or unidirectional causality. It means that these markets are interconnected. In addition, the analysis shows limited or no opportunities for arbitrage from daily price differentials. However, the reduction in the number of statistically significant causal relationship for prices that are a week or month apart may imply increased opportunities for arbitrage. The extent to which these markets are interconnected or integrated is investigated in the next section.

5.3. Results of cointegration tests

This study will present three kinds of cointegration tests: (i) testing for cointegration among regional markets, (ii) testing for

cointegration between each regional market and Henry Hub, and (iii) testing for cointegration between spot and futures markets. The result of testing for cointegration between regional markets is presented in Table 4.

The table provides empirical results on the number of cointegrating relationships between the eight regional markets studied in this research. The results clearly demonstrate that there are seven cointegrating equation(s) at the 5% level of significance. That is, all the regional markets are integrated or cointegrated such that prices observed in one market tend to move in the same direction as prices in other markets. Supply would be channeled to the market that commands the highest price. The implication of this finding is that these regional markets are interrelated, and that the law of one price may be operative. In other words, in the long run all the regional markets could exhibit one market-clearing (equilibrium) price after allowing for transportation and transaction costs.

Table 4
Cointegration between regional markets.^a

Hypothesized number of cointegrating relations	Eigenvalue	Trace statistic	5 Percent critical value	Prob.**
$H_0: r = 0$	0.166	1124.73	159.53	0.000*
$H_0: r \leq 1$	0.107	701.93	125.62	0.000*
$H_0: r \leq 2$	0.079	437.99	95.75	0.000*
$H_0: r \leq 3$	0.061	245.36	69.82	0.000*
$H_0: r \leq 4$	0.019	99.91	47.86	0.000*
$H_0: r \leq 5$	0.013	55.12	29.80	0.000*
$H_0: r \leq 6$	0.008	25.01	15.49	0.001*
$H_0: r \leq 7$	0.002	5.34	3.84	0.021*

*(**) Denotes rejection of the hypothesis at the 5%(1%) level.

Trace test indicates 8 cointegrating equation(s) at the 0.05 level.

* denotes rejection of the hypothesis at the 0.05 level.

**MacKinnon–Haug–Michelis (1999) *p*-values [39].

^a Markets studied were Henry, AECO, Kern, Kingsgate, Malin, WYOMING, Stanfield and Sumas.

Henry Hub is the most liquid of all markets. Prices in other market hubs tend to follow prices at Henry Hub. Therefore, cointegration between prices in other hubs and that of prices at Henry Hub was investigated. The results are presented in Table 5. The results show that the null hypothesis of no cointegrating relations is rejected for all markets. However, the null hypothesis of no more than one cointegrating relations is not rejected. The implication of this finding is that there is a long run relationship between prices in other markets and that of Henry Hub, which prevent divergence. The results also imply that these markets exhibit autoregressive process of order one (AR(1)) and that they will exhibit stationarity after differencing once.

The study also undertook another cointegration analysis in which the objective was to understand how spot prices observed at several regional markets or hubs move compared with the futures price (NYMEX). The results are presented in Table 6. The results of tests of cointegration between spot prices at various market and the futures price at NYMEX show that (i) there is a cointegrating relationship between regional spot and futures prices, and (ii) the number of cointegrating relations is no more than one. Thus, the relationship between spot and futures price is characterized by the regression of the former (spot) on the latter (future) resulting in a coefficient of one. In the long-run relationship, therefore, spot price equals future price plus a premium for risk.

5.4. Risk premium, rationality and market efficiency

The result of testing for a risk premium ($\alpha_i = 0$), rationality of futures market ($\beta_i = 1$), and joint test of rationality and risk premium constancy or market efficiency ($\alpha_i = 0$ and $\beta_i = 1$), are presented in Table 7. The null hypothesis of risk premium constancy or risk neutrality ($H_0: \alpha_i = 0$) was rejected at 1 percent level of probability in all cases except weeks 1–3 (Table 7). This means that for contracts with about a month to maturity, market participants require low risk premium (negligible difference between spot and futures price) because they do not exhibit a high degree of risk aversion or risk premium is relatively constant. The low risk premium or risk-taking behavior for contracts with less than 4 weeks to maturity could be because (i) market participants have limited opportunity to hedge, change or modify contracts, (ii)

Table 5
Cointegration between regional markets and Henry Hub, 1996–2004.

Market pairs	Hypothesized No. of CE(s)	Eigenvalue	Trace statistic	Prob.**
Henry Hub and Aeco	$H_0: r = 0$	0.025	91.35	0.000*
	$H_0: r \leq 1$	0.003	10.83	0.094
Henry Hub and Kern	$H_0: r = 0$	0.017	64.64	0.000*
	$H_0: r \leq 1$	0.003	10.43	0.109
Henry Hub and Kingsgate	$H_0: r = 0$	0.033	89.58	0.000*
	$H_0: r \leq 1$	0.001	1.85	0.205
Henry Hub and Malin	$H_0: r = 0$	0.033	97.11	0.000*
	$H_0: r \leq 1$	0.000	0.92	0.390
Henry Hub and Wyoming	$H_0: r = 0$	0.017	65.13	0.000*
	$H_0: r \leq 1$	0.004	10.99	0.089
Henry Hub and Stanfield	$H_0: r = 0$	0.034	109.71	0.000*
	$H_0: r \leq 1$	0.001	2.20	0.162
Henry Hub and Sumas	$H_0: r = 0$	0.037	121.80	0.000*
	$H_0: r \leq 1$	0.001	2.28	0.155

Trace test indicates 1 cointegrating equation(s) at the 0.05 level.

* Denotes rejection of the hypothesis at the 0.05 level. ** Denotes rejection of the hypothesis at the 5% level (the 5% critical values for $H_0: r = 0$ and $H_0: r < 1$ are 12.32 and 4.13, respectively).

**MacKinnon–Haug–Michelis (1999) *p*-values [39].

Table 6
Cointegration between regional spot prices and futures price.

Market pairs	Hypothesized No. of CE(s)	Eigenvalue	Trace statistic	Prob.**
Henry Hub and monthly forward	$H_0: r = 0$	0.257	20.32	0.009*
	$H_0: r \leq 1$	0.063	3.65	0.056
Aeco and monthly forward	$H_0: r = 0$	0.217	17.20	0.028*
	$H_0: r \leq 1$	0.061	3.50	0.061
Kern and monthly forward	$H_0: r = 0$	0.125	10.83	0.222
	$H_0: r \leq 1$	0.059	3.38	0.066
Kingsgate and monthly forward	$H_0: r = 0$	0.174	14.56	0.069
	$H_0: r \leq 1$	0.080	4.40	0.036*
Malin and monthly forward	$H_0: r = 0$	0.170	11.39	0.189
	$H_0: r \leq 1$	0.028	1.49	0.223
Wyoming and monthly forward	$H_0: r = 0$	0.128	11.97	0.158
	$H_0: r \leq 1$	0.085	4.71	0.030*
Stanfield and monthly forward	$H_0: r = 0$	0.222	17.08	0.029*
	$H_0: r \leq 1$	0.058	3.29	0.070
Sumas and monthly forward	$H_0: r = 0$	0.257	20.32	0.009*
	$H_0: r \leq 1$	0.063	3.65	0.056

* Denotes rejection of the hypothesis at the 0.05 level.

Note: futures prices are quotes from the NYMEX.

** Denotes rejection of the hypothesis at the 5% level (the 5% critical values for $H_0: r = 0$ and $H_0: r < 1$ are 15.49 and 3.84, respectively).

limited opportunities to make storage and transportation-related adjustments, and (iii) minimal value of market information to earn profit.

For weeks 4–75, risk neutrality or constancy was rejected at 1% and 5% level of significance. That is market participants become risk averse, apply higher transportation costs, or require higher premiums. Several factors may have contributed to these results. First, the characteristic of natural gas is such that a minimum volume of about 10,000 Decatherm (Dth) is required to execute a transaction, thus rendering this commodity to be less amenable for purchases in smaller increments (indivisibility).⁴ There are limited natural gas storage and transportation (pipeline) resources. In addition to these characteristics of the natural gas market, factors such as (i) international forces (e.g., political disruption in oil producing regions); (iii) weather related variables (e.g., as hurricane); (iv) lack of transparency and asymmetry of information regarding the accuracy of market prices; and (v) unregulated activities of hedge funds, brokers and banks, may have contributed toward extreme volatility of natural gas production, supply and prices [40,41]. Because of these factors, it is likely that market participants tend to exhibit risk-averse behavior or require higher premium for contracts with four to seventy-five weeks to maturity. These and related factors have compounded the ability of experts to make reasonable or accurate forecasts regarding the future spot price of natural gas.

Rationality of future market price implies that its coefficient (β_i) should be close to unity ($\beta_i \sim 1$). The findings of this study indicate that rationality of future market price is observed only in the last three weeks before contract maturity. This means the value or usefulness of futures prices for planning short and long-term plans becomes negligible.

The joint test of risk neutrality and efficiency of the futures market was rejected for all times periods considered except during

⁴ Please see http://www.nymex.com/ng_pre_agree.aspx.

Table 7
Results of tests for risk neutrality.

Weeks to contract maturity	Test for risk neutrality ($\alpha = 0$)	Test for rationality ($\beta = 1$)	Joint test of efficiency ($\beta = 1, \alpha = 0$)	R^2	Weeks to contract maturity	Test for risk neutrality ($\alpha = 0$)	Test for rationality ($\beta = 1$)	Joint test of efficiency ($\beta = 1, \alpha = 0$)	R^2
Week1	0.025	0.945	5.27	0.92	Week39	0.418*	0.167*	14.63*	0.90
Week2	0.037	0.917	4.16	0.92	Week40	0.392*	0.218*	13.37*	0.90
Week3	0.059	0.867	5.54	0.91	Week41	0.466*	0.082*	17.41*	0.90
Week4	0.404*	0.151*	138.80*	0.90	Week42	0.410*	0.185*	10.34*	0.90
Week5	0.444*	0.085*	184.58*	0.90	Week43	0.458*	0.097*	13.75*	0.90
Week6	0.451*	0.075*	139.48*	0.90	Week44	0.436*	0.140*	11.06*	0.90
Week7	0.479*	0.032*	126.93*	0.90	Week45	0.476*	0.066*	12.95*	0.90
Week8	0.505*	-0.006*	134.54*	0.90	Week46	0.531*	-0.026*	21.04*	0.90
Week9	0.505*	-0.006*	135.05*	0.90	Week47	0.547*	-0.051*	17.42*	0.90
Week10	0.534*	-0.048*	89.38*	0.90	Week48	0.552*	-0.057*	18.23*	0.89
Week11	0.592*	-0.127*	85.21*	0.90	Week49	0.515*	0.006*	15.96*	0.89
Week12	0.580*	-0.113*	56.23*	0.90	Week50	0.582*	-0.106*	17.91*	0.89
Week13	0.528*	-0.042*	33.53*	0.90	Week51	0.699*	-0.274*	23.08*	0.89
Week14	0.475*	0.040*	14.77*	0.90	Week52	0.894*	-0.511*	27.62*	0.89
Week15	0.427*	0.121*	6.63*	0.90	Week53	0.935*	-0.549*	44.68*	0.89
Week16	0.425*	0.124*	11.85*	0.90	Week54	0.783*	-0.381*	34.09*	0.89
Week17	0.409*	0.152*	15.62*	0.90	Week55	0.661*	-0.213*	24.38*	0.89
Week18	0.454*	0.076*	24.04*	0.90	Week56	0.578*	-0.096*	18.00*	0.89
Week19	0.458*	0.068*	31.07*	0.90	Week57	0.495*	0.054*	11.72*	0.89
Week20	0.453*	0.078*	33.56*	0.90	Week58	0.527*	0.006*	12.45*	0.89
Week21	0.438*	0.101*	19.98*	0.90	Week59	0.453*	0.133*	9.18*	0.89
Week22	0.475*	0.042*	36.59*	0.90	Week60	0.456*	0.134*	11.11*	0.89
Week23	0.451*	0.081*	26.50*	0.90	Week61	0.382*	0.280*	8.52*	0.89
Week24	0.415*	0.142*	17.40*	0.90	Week62	0.333*	0.379*	6.53*	0.89
Week25	0.349*	0.267*	12.61*	0.90	Week63	0.330*	0.385*	3.96	0.89
Week26	0.338*	0.291*	9.58*	0.90	Week64	0.373*	0.300*	4.83	0.89
Week27	0.301*	0.367*	5.89	0.91	Week65	0.408*	0.234*	4.39	0.89
Week28	0.312*	0.347*	5.79	0.91	Week66	0.486*	0.085*	5.99*	0.89
Week29	0.362*	0.246*	8.99*	0.90	Week67	0.454*	0.145*	4.34	0.89
Week30	0.416*	0.144*	12.95*	0.90	Week68	0.457*	0.137*	6.48*	0.89
Week31	0.428*	0.124*	18.84*	0.90	Week69	0.498*	0.083*	8.70*	0.89
Week32	0.455*	0.077*	19.50*	0.90	Week70	0.483*	0.112*	8.85*	0.89
Week33	0.513*	-0.019*	27.90*	0.90	Week71	0.495*	0.089*	12.54*	0.89
Week34	0.576*	-0.115*	50.45*	0.90	Week72	0.528*	0.034*	14.34*	0.88
Week35	0.566*	-0.102*	50.58*	0.90	Week73	0.605*	-0.056*	12.43*	0.88
Week36	0.553*	-0.083*	46.09*	0.90	Week74	0.578*	-0.024*	14.12*	0.88
Week37	0.486*	0.025*	32.35*	0.90	Week75	0.572*	-0.014*	13.06*	0.88
Week38	0.415*	0.173*	16.72*	0.90					

* Denotes significance at 1% and 5% levels. The regression is based on logarithm of prices.

the last three weeks to maturity.⁵ The failure of the joint test implies that the natural gas futures price cannot be concluded to be an unbiased predictor of prices at delivery. As seen in recent years, not all market information related to natural gas production and marketing is publicly available [38,39]. In addition, the market price data reported may not always be an accurate reflection of actual transactions that took place. Due to these and factors discussed earlier, the information set is neither “accurate” nor “transparent”. These market environments are not conducive for the realization of market efficiency. If futures markets were to be used for planning the acquisition of least cost resources and enhance the reliability of energy delivery, it is necessary to make sure that data on production, supply, imports, exports, and prices at different market hubs and different transactions become accurate and publicly available. In addition, measures that minimizes the impacts of domestic, international and natural (weather related) factors on production, supply, and prices should be identified and implemented.

⁵ There are three forms of efficiency: weak, semi-strong and strong. All of them refer to the unbiasedness of futures market prices as a forecast for spot prices, but with different sets of information. The weak form relies on the sequences of past prices, the semi-strong on past prices and publicly available information, and the strong form on all relevant information.

5.5. Constancy of risk and rationality of regional market hubs

Henry hub is the most liquid market and relatively better predicted by futures prices. This study performed analysis of the impact of sequentially adding different length of average futures prices to find out changes in risk premium and rationality of futures prices, or assess improvements to the predictability of spot prices (see Table 8). As it can be seen from the table, sequentially adding average futures price of various length, (i) increased the intercept term (measure of risk premium, transportation, transaction costs, and so on) from 13 to 19 cents per MMBtu; (ii) the coefficient remains about unity, while becoming small or negative for other months; and (iii) there is no substantial improvement in R-square

Table 8
Explanatory power of periodic average of futures prices on Henry Hub spot prices.

Intercept	Estimates of coefficient on forward price						R^2
	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	
0.131*	0.950*						0.959
0.156*	1.070*	-0.128*					0.96
0.149*	1.075*	-0.156**	0.026				0.96
0.172*	1.090*	-0.198*	0.190**	-0.146**			0.961
0.184*	1.081*	-0.187**	0.155	0.016	-0.135		0.961
0.188*	1.078*	-0.184**	0.155	-0.002	-0.061	-0.059	0.961

* and ** denotes significance at 1% and 5% levels, respectively.

as more futures prices were added to the analysis. This finding seems to confirm the fact that futures market are efficient only for about a month to maturity regardless of the amount of information contained in futures prices of various length.

6. Conclusions and recommendations

Natural gas provides about a quarter of the nation's energy portfolio. The emphasis on natural gas is good news for the economy, the environment, and society because it is relatively cheap compared with other conventional fuels, and it is less damaging to the environment. Natural gas has played a vital role in meeting almost a quarter of the nation's energy portfolio. It is expected that natural gas could be a growing source of energy to power our economy for many years to come.

It is argued that the U.S. has enough natural gas to meet the increased demand for natural gas in the foreseeable future. However, it has become increasingly important to insure consistent supply to minimize the impact of price volatility on residential and commercial/industrial customers. Furthermore, it is essential to minimize the impacts of interruption in natural gas supply and natural gas price hikes because it could produce significant repetitive economic effect throughout the economy.

High and volatile prices are often disruptive but are claimed to be self-correcting. That is, higher prices discourage consumption growth and encourage supply growth, thereby relieving the short-fall that caused the price spike. A boom in the natural gas drilling activities in response to price spikes indicates self-corrective actions. However, the ability of this market to self-correct come after the economy passes through a series of hardships. Thus, despite the claim of being self-correcting, the natural gas market is considered risky.

The natural gas sector is characterized by a high degree of risk due to weather, market prices, transportation or capacity, storage, and demand factors. To extract the most economic value from this risky yet extremely important fuel, it is essential to study selected key attributes of the market. This study examined causality, cointegration, and efficiency of the natural gas market.

The present study implemented several kinds of statistical analyses using spot and futures market data. The conclusions from the present study are that:

- (i) Spot and futures prices of natural gas are linked or move in the same direction. That is, the variance between spot and futures prices declines over time. The long-run relationship, therefore, would be spot price equals futures price plus premium for risk, where the risk premium could be zero;
- (ii) Spot prices tend to cause the movement of futures prices for contracts with less than one year to maturity in 50% of the cases; while future price exert significant impact in 50% of the case for contract that are more than 1–2 year to maturity, and statistically significant for contracts that are 3 or less months to maturity.
- (iii) Spot prices in the market hubs studied exhibit a bi-directional or two-way causal relationship, suggesting instantaneous response of price changes across markets;
- (iv) Market hubs located close to major sources of supply (AECO, Kingsgate, Sumas, Henry Hub) tend to exert statistically significant impacts on movements of prices in other regional markets, and this result holds true whether the analysis uses daily, weekly or monthly data;
- (v) Arbitrage opportunities may not exist from daily market price differentials. However, regional market prices that are weeks or months apart may offer opportunities to acquire least cost resources or profit from arbitrage.

- (vi) All the regional markets are integrated or cointegrated such that prices observed in one market tend to move in the same direction as prices in other locations. Thus, in the long run, all the regional markets could exhibit one market-clearing (equilibrium) price (the law of one price) after allowing for transportation and transaction costs;
- (vii) All regional market prices tend to move in the direction of prices observed at Henry Hub, thus maintaining a constant variance between prices at Henry Hub and other markets;
- (viii) Market participants demand a higher risk premium for contracts that are 4–75 weeks to maturity. This could be because of a relatively longer period required to close discrepancies between supply and demand, limited and inaccurate market information, or market manipulation that resulted in irrational market behavior. On the other hand, the study showed an insignificant or negative premium for contracts that are three weeks or less, and for contracts that are more than one year to maturity; and
- (ix) The natural gas market is not efficient if forecasts of futures market prices are based on only past spot prices. That is, it may be necessary to study variables other than past spot prices that may influence the differences between spot and futures prices.

Acknowledgements

This study was completed during the author's tenure with the Washington Utilities and Transportation Commission, Olympia, Washington. The author appreciates the support provided by WUTC and colleagues. The views expressed and the contents of this paper are solely the responsibility of the author and not that of the institution with which he is affiliated.

References

- [1] EIA. Annual Energy Outlook 2003 with Projections to 2025. Report #: DOE/EIA-0383, January, 2003.
- [2] <http://www.naturalgas.org/business/industry.asp>; 2010.
- [3] Henning Bruce, Sloan M, Leon MD. Natural gas and energy price volatility. Energy and Environmental Analysis, Inc. Prepared for the Oak Ridge National Laboratory; October, 2003.
- [4] NGSA, Natural Gas Supply Association, <http://www.naturalgas.org/regulation/regulation.asp>; 2010.
- [5] Suenaga H, Smith A, Williams J. Volatility dynamics of NYMEX natural gas futures prices. Working Paper Series 2007-05, Curtin Business School, Australia, 2007.
- [6] Walls DW. Competition, prices, and efficiency in the deregulated gas pipeline network: a multivariate cointegration analysis. *Journal of Energy & Development* 1995;19:1–14.
- [7] De Vany AS. The law of one price in a network: arbitrage and price dynamics in natural gas city gate markets. *Journal of Regional Science* 1996;36:555–70.
- [8] Doane, Michael J, Spulber D. Open access and the evolution of the U.S. spot market for natural gas. *Journal of Law and Economics* 1994;37:477–517.
- [9] Cuddington JT, Wang Z. Statistical analyses of the geographic market delineation with an application to the U.S. natural gas markets. GU working paper #99-12, October, 1999.
- [10] <http://www.futurestech.net/history.htm>.
- [11] <http://rru.worldbank.org/documents/publicpolicyjournal/141juris.pdf>.
- [12] Chinn M, LeBlanc M, Coibion O. The predictive characteristics of energy futures: recent evidence for crude oil, natural gas, gasoline, and heating oil. UCSC Economics Working Paper No. 490, 2001.
- [13] Cuddington J, Wang Z. Assessing the degree of spot market integration for U.S. natural gas: evidence from daily price data. *Journal of Regulatory Economics* 2006;29:195–210.
- [14] De Vany AS, Walls D. Pipeline access and market integration in the natural gas industry: evidence from cointegration tests. *The Energy Journal* 1993;14:1–19.
- [15] Walls DW. A cointegration rank test of market linkages with an application to the U.S. natural gas industry. *Review of Industrial Organization* 1994;9:181–91.
- [16] Walls DW. Price convergence across natural gas fields and city markets. *The Energy Journal* 1994;15:37–49.
- [17] Serletis A. Is there an east-west split in north-american natural gas markets? *The Energy Journal* 1997;18:47–62.

- [18] Phillips PCB, Perron P. Testing for a unit root in time series regressions. *Biometrika* 1988;75:335–46.
- [19] Dickey DA, Fuller W. Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association* 1979;74:427–31.
- [20] King M, Cuc M. Price convergence in North American natural gas spot markets. *The Energy Journal* 1996;17:17–42.
- [21] Granger CWJ. Investigating causal relationship by econometric models and cross spectral methods. *Econometrica* 1969;37:424–38.
- [22] Zestos GK, Tao X. Trade and GDP growth: causal relations in the United States and Canada. *Southern Economic Journal* 2002;68:859–74.
- [23] De Vany AS. Natural gas industry transformation, competitive institutions and the role of regulation: lessons from open access in US natural gas markets. *Energy Policy* 1994;22:755–63.
- [24] Working H. Futures trading and hedging. *American Economic Review* 1953;43:314–43.
- [25] Raynauld J, Tessier J. Risk premiums in futures markets: an empirical investigation. *Journal of Futures Markets* 1984;4:189–211.
- [26] Newbery D, Stiglitz MG. *The theory of commodity price stabilization*. Oxford: Clarendon Press; 1981.
- [27] Brenner R, Kroner KF. Arbitrage, cointegration, and testing the unbiasedness hypothesis in financial markets. *Journal of Financial and Quantitative Analysis* 1995;30:23–42.
- [28] Engle RF, Granger CWJ. Cointegration and error correction: representation, estimation, and testing. *Econometrica* 1987;55:251–76.
- [29] Russell PS, Torbey VM. The efficient market hypothesis on trial: a survey, <http://www.westga.edu/~bquest/2002/market.htm>; 2002.
- [30] Serletis A, Rangel-Ruiz R. Testing for common features in North American energy markets. *Energy Economics* 2004;26:401–14.
- [31] Johansen S. Estimation and hypothesis testing of cointegration vectors in Gaussian vector autoregressive models. *Econometrica* 1991;59:1551–80.
- [32] Johansen S. Likelihood-based inference in cointegrated vector autoregressive models.
- [33] Phillips PCB. Time series regression with a unit root. *Econometrica* 1987;55:277–301. Oxford: Oxford University Press, 1995.
- [34] Fama E. Efficient capital markets: a review of theory and empirical work. *The Journal of Finance* 1970;25:383–417.
- [35] Herbert J. The relation of monthly spot to futures prices for natural gas. *Energy* 1993;18:1119–24.
- [36] Herbert J. Trading volume, maturity, and natural gas futures price volatility. *Energy Economics* 1995;17:293–9.
- [37] Rajaraman I. Testing the rationality of futures prices for selected least developed countries agricultural export. *Journal of Futures Markets* 1986;6:523–40.
- [38] MacKinnon J. Critical values for cointegration tests. In: Engle RF, Granger CWJ, editors. *Long run economic relationships*. Oxford: Oxford University Press; 1991. p. 269–76.
- [39] MacKinnon J, Haug A, Michelis L. Numerical distribution functions of likelihood ratio tests for cointegration. *Journal of Applied Econometrics* 1999;14:563–77.
- [40] Cooper NM. The role of supply, demand and financial commodity markets in the natural gas price spiral. Prepared for Midwest attorneys general natural gas working group (Illinois, Iowa, Missouri, Wisconsin), March, 2006.
- [41] United States Government Accountability Office (US GAO). Natural gas and electricity markets: Federal Government Actions to Improve Private Price Indices and Stakeholder Reaction. Report to the Ranking Minority Member, Committee on Energy and Natural Resources, U.S. Senate, December, 2005.

Yohannes Kebede Gebre-Mariam (AKA Yohannes Kebede or Yohannes Mariam) holds B.Sc. in Agricultural Economics from Addis Ababa University (Alemaya), Ethiopia; M.Sc. in Agricultural Economics/Economics and Ph.D. in multidisciplinary program involving Agricultural Economics, Economics, Anthropology and Applied Cognitive Psychology from McGill University, Montreal, Canada. He has worked as Environmental Economist/Social Scientist for Environment Canada, and Senior Economist for Washington Utilities and Transportation Commission. Presently, he works as a Senior Economist for the Office of the People's Counsel for the District of Columbia. He is also an Adjunct Professor of Economics Management with the University of Phoenix and Florida Keys Community College. This study was completed during the author's tenure with the Washington Utilities and Transportation Commission, Olympia, Washington. The views expressed and the contents of this paper are solely the responsibility of the author and not that of the institution with which he is affiliated.